CSAW Hack3D 2026 Qualifier Report

Team: No Fix
Red Team Vs Blue Team - .PLY Me a Secret
October 4, 2026 (public copy)

1. Our approach

We worked with Boat.ply, Box.ply, and Tower.ply. Our goal was to recover the main shapes, find the material clues, and produce files that could be opened again. We used a small Python script and basic geometric shapes. We kept the reconstruction simple and recorded the parts where the evidence was incomplete.

2. Reading the files

We first read the PLY files as text instead of only opening them in a model viewer. Each file had a URL hidden in it. Boat and Tower had extra text after a vertex row, and Box had a link after end_header. We kept the first three numbers as x, y, and z and read the triangle indices separately. We preserved the original files, including the RGB values in Box.

File	Vertices	Faces	Clue location
Boat.ply	141,827	280,629	Line 23: 3DBenchy website
Box.ply	338,375	666,248	Line 12: Link to challenge archive
Tower.ply	30,118	59,872	Line 19: paper DOI and Phi = 40%

3. Following the clues

For Boat, we followed the 3DBenchy link and found the official model and its dimension page [1]. The hull, cabin, and chimney matched the scan, so we used the official geometry to recover the missing details.

For Box, the header link led to a challenge archive with a password hint. The hint pointed to the conference name. After opening the archive, we identified a paper about cenosphere/HDPE syntactic foams [2]. Figure 12, part 4, showed an electrical connector with the same four-port shape.

For Tower, we followed the DOI in the vertex row [3]. The paper describes hollow glass microspheres in an acrylate-based resin. We interpreted the accompanying Phi = 40% as the filler volume fraction, consistent with the related materials dataset [4].

4. Rebuilding the parts

Boat. We used the official single-part 3MF as the reference. We converted it to STL with Trimesh and Manifold, which cleaned very thin triangles in the reference mesh. The restored boat has the normal Benchy dimensions: about 60 x 31 x 48 mm. We include the reference 3MF as well as our STL. This is a reconstruction from an identified public model, rather than a new design of our own.

Tower. We used PCA to place the longest direction of the scan along the cylinder axis. We fitted a circle to the middle of the side wall and used the median positions of the two end regions for the height. The estimates were 12.489 mm in diameter and 24.237 mm in height. We rounded them to 12.5 mm and 24.2 mm, then made a cylinder with flat end caps.

Box. We inspected slices through the scan and approximated it with a round cup and four side tubes. We used a 61.3 mm outside body diameter, a 57.3 mm inside diameter, and a 27 mm height. The overall spans are 98 mm and 92 mm. We added two simple mounting bosses. We used smooth 20 mm bores because the scan did not give enough detail to rebuild the threads reliably.

We treated the scan coordinates as millimeters. The Boat scan was consistent with the 60 mm Benchy reference, but the PLY files did not explicitly state a unit. The Box and Tower dimensions should therefore be read as scan-based estimates.

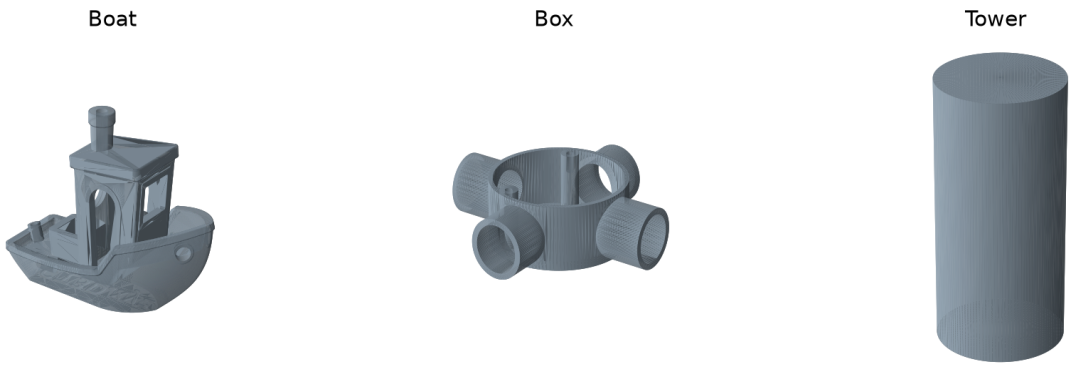


Figure 1. Our three reconstructed models. The detailed views and dimensions are on the three drawing sheets at the end of this report.

The Box reconstruction keeps the main features and has approximate dimensions. Fine threads, small fillets, and exact screw locations would need better measurements before manufacturing.

5. Materials

We separated what we could recover from what we selected for a basic reproduction. A PLY surface scan by itself does not identify a polymer or an exact mixture.

Part	Composition and evidence
Boat	We selected PLA (polylactic acid) for reproduction. The original scanned material is undetermined. An official Benchy printing example uses PLA [5], but that does not prove that our scan used it.
Box	We identified the material family as HDPE with fly-ash cenospheres, based on the matching connector in [2]. The paper studies 20, 40, and 60 wt.% filler; it does not uniquely label this connector's loading. For reproduction, we propose 80 wt.% HDPE + 20 wt.% cenospheres. This is a selected recipe, not a recovered exact ratio.
Tower	We infer 60 vol.% acrylate-based photopolymer resin + 40 vol.% hollow glass microspheres from [3] and the Phi clue. These are volume percentages. The particle density grade and exact resin formulation are undetermined.

6. Checks and limitations

We exported each STL and loaded the saved file again. All three saved meshes were closed and had consistent face winding. We also checked that they had positive volume and the expected overall size. These checks verify the digital files; we have not physically printed or tested the parts.

Part	STL triangles	Saved overall size (mm)	Closed mesh
Boat	225,146	60.001 x 31.004 x 48.000	Yes
Box	7,492	98.0 x 92.0 x 27.0	Yes
Tower	512	12.5 x 12.5 x 24.2	Yes

We include the reconstruction script, its dependencies, and the public Benchy reference. The code can recreate the models using our recorded dimensions; the original organizer scans are not included in this public copy. The main remaining uncertainties are Box's fine details and filler loading, and Boat's original material.

References

[1] Creative Tools, 3DBenchy. Official dimensions; official model repository. Accessed October 4, 2026.

[2] B. R. Bharath Kumar et al., Processing of cenosphere/HDPE syntactic foams using an industrial scale polymer injection molding machine, Materials and Design 92 (2016), 414-423. doi:10.1016/j.matdes.2015.12.052. Supplied in the challenge archive; see section 2.1, Figure 12, and Table 6.

[3] C. Beckwith et al., Rheology guided additive manufacturing of thermosetting syntactic foams: mechanical behavior and failure mechanisms, Composites Part A 199 (2025), 109248. doi:10.1016/j.compositesa.2025.109248. We used the public abstract and the clue in Tower.ply.

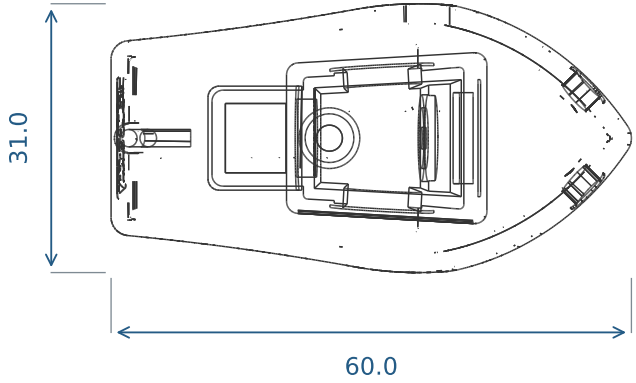
[4] C. Beckwith and N. Gupta, Rheological Measurement Dataset of Resin and Composite Mixtures for Digital Light Processing based Additive Manufacturing, Scientific Data 13 (2026), 1043. doi:10.1038/s41597-026-07391-0.

[5] 3DBenchy, 3D-printing on MakerBot Replicator 2, PLA printing example. Accessed October 4, 2026.

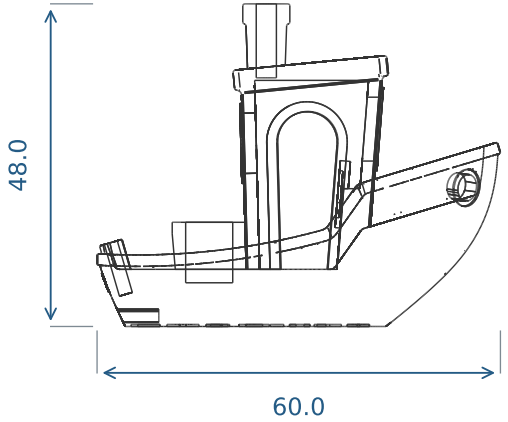
No Fix - Boat reconstruction

Dimensions in mm | Not to scale

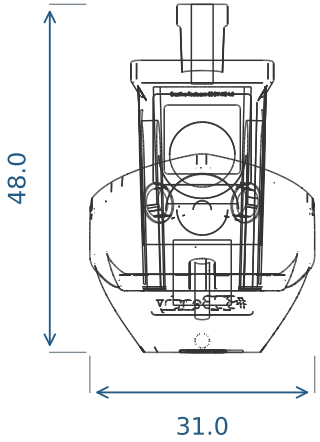
Top view



Side view



End view



Isometric view

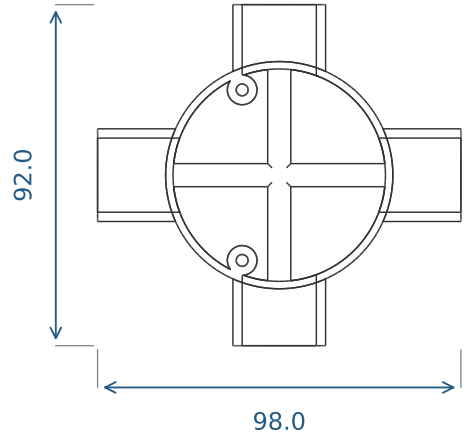


Nominal dimensions from 3DBenchy: length 60.00, width 31.00, height 48.00 mm.
Chimney: outer diameter 7.00 mm; bore 3.00 mm. Roof length: 23.00 mm.
Recovered from the official Creative Tools model. Dimensions are reference values.

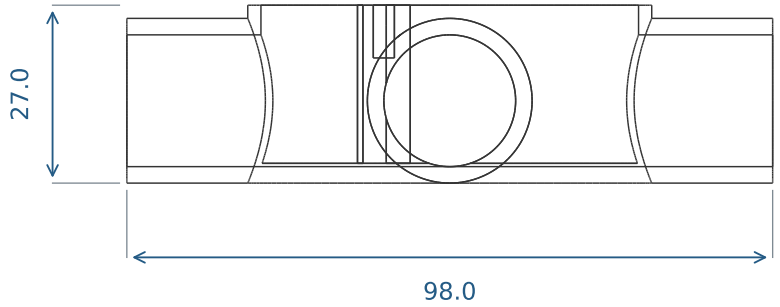
No Fix - Box reconstruction

Dimensions in mm | Not to scale

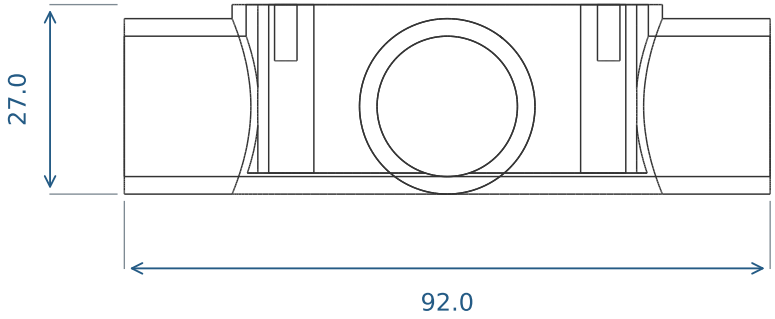
Top view



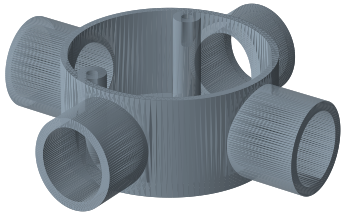
Side view



End view



Isometric view

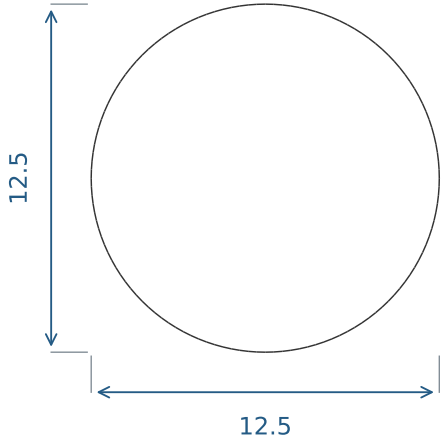


Approximate scan-based reconstruction: body OD 61.3, ID 57.3 mm; wall 2.0 mm; floor 3.0 mm.
Four ports: OD 25.0, bore 20.0 mm. Two approximate bosses: OD 8.0; blind holes 3.2 mm.
Smooth bores replace unresolved thread detail. Dimensions are estimates, not manufacturing tolerances.

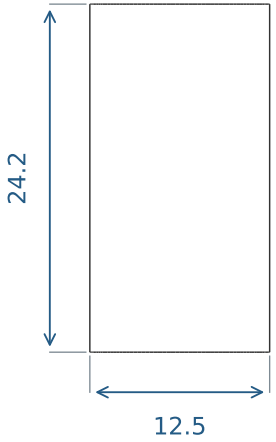
No Fix - Tower reconstruction

Dimensions in mm | Not to scale

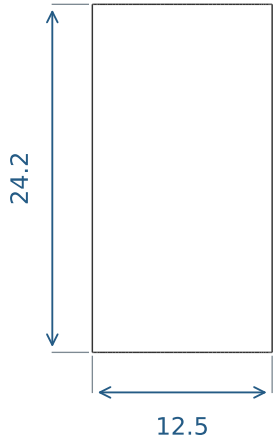
Top view



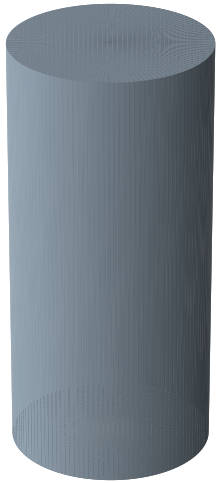
Side view



End view



Isometric view



Cylinder fitted to the scan: diameter 12.5 mm; height 24.2 mm.
Unrounded estimates: diameter 12.489 mm; height 24.237 mm.
The model uses planar end caps. Dimensions are estimates from the supplied scan.